Numerical Study of Heating During Hypersonic Descent Inspired by the X-43

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This study examines aerodynamic heating during a simplified hypersonic descent inspired by the NASA X-43. A five-state initial-value problem describes altitude, speed, flight-path angle, stagnation-region temperature, and constant vehicle mass. The model includes altitude-dependent atmospheric density and gravity, constant lift and drag coefficients, convective heating, and radiative cooling. The equations are integrated using a classical fourth-order Runge–Kutta method implemented from scratch. Timestep refinement demonstrates convergence toward fourth-order accuracy, while exact special-case solutions, independent derivative benchmarks, and a mechanical energy identity check the implementation. The baseline temperature peaks at 2632.82 K**. Increasing initial speed from** 1700 **to** 2500 m/s **raises peak temperature from** 2373.43 **to** 2962.27 K**, whereas increasing emissivity from** 0.5 **to** 1.0 **reduces it from** 3006.14 **to** 2528.01 K**. These predictions describe the assumed model over a** 400 s **observation interval rather than a validated reconstruction of an X-43 flight.**

I. Nomenclature

A	=	reference area, m^2
C_A	=	effective heat capacity per unit area, $\mathrm{J}/(\mathrm{m}^2\,\mathrm{K})$
C_D, C_L	=	drag and lift coefficients
c_p	=	specific heat capacity, $\mathrm{J}/(\mathrm{kg\,K})$
D, L	=	drag and lift forces, N
E	=	specific mechanical energy, J/kg
g, g_0	=	local and surface gravitational accelerations, $\mathrm{m/s}^2$
H	=	atmospheric scale height, m
h, h_0	=	altitude and baseline initial altitude, m
K	=	convective-heating coefficient, $\mathrm{kg}^{1/2}/\mathrm{m}$
m	=	vehicle mass, kg
R_E	=	Earth radius, m
R_n	=	nose radius, m

T_s = approximate stagnation-region temperature, K
V = vehicle speed, m/s
γ = flight-path angle, rad
δ = effective heated thickness, m
ϵ = surface emissivity
ρ_a, ρ_{30} = local and reference atmospheric densities, $\mathrm{kg/m^3}$
ρ_s = material density, $\mathrm{kg/m^3}$
σ = Stefan–Boltzmann constant, $\mathrm{W/(m^2\,K^4)}$
$\dot{q}_{\mathrm{conv}}$ = convective heat flux, $\mathrm{W/m^2}$
Q_{conv} = integrated convective heat load per unit area, $\mathrm{J/m^2}$
Δt = numerical timestep, s

II. Introduction

Hypersonic vehicles experience substantial aerodynamic heating, and their thermal response depends on both the trajectory and the balance between convective heating and radiative cooling. Inspired by the NASA X-43 program [1], this study considers an unpowered vehicle beginning at 30 km with speed 2112.14 m/s, approximately Mach 7 under the assumed initial conditions, flight-path angle $-2°$, and mass 1400 kg. The vehicle and material parameters are modeling assumptions rather than verified X-43 flight data.

The primary questions are how initial speed affects convective heating and peak temperature, and how emissivity affects the thermal response. A supplementary study varies the initial flight-path angle. Addressing these questions requires tracking how altitude-dependent density changes aerodynamic forces and heating while the vehicle speed and trajectory evolve. These nonlinear relationships, together with fourth-power radiative cooling, produce a coupled initial-value problem without a known closed-form solution for the full model. We therefore use numerical integration and assess its accuracy through timestep refinement and independent implementation checks.

III. Mathematical Model

The vehicle is represented by a point mass with locally flat flight geometry. Earth rotation and trajectory-curvature terms are neglected, while gravity is allowed to vary with altitude. The state and initial-value problem are

$$\mathbf{y} = [h,\ V,\ \gamma,\ T_s,\ m]^T, \qquad \frac{d\mathbf{y}}{dt} = \mathbf{f}(t, \mathbf{y}), \qquad \mathbf{y}(0) = \mathbf{y}_0, \tag{1}$$

with baseline initial conditions

$$\mathbf{y}_0 = [30000,\ 2112.14,\ -2\pi/180,\ 300,\ 1400]^T. \tag{2}$$

The components use meters, meters per second, radians, kelvin, and kilograms, respectively. A negative flight-path angle denotes descent.

A. Flight Dynamics

The translational equations are

$$\frac{dh}{dt} = V \sin\gamma, \tag{3}$$

$$\frac{dV}{dt} = -\frac{D}{m} - g(h)\sin\gamma, \tag{4}$$

$$\frac{d\gamma}{dt} = \frac{L}{mV} - \frac{g(h)\cos\gamma}{V}, \tag{5}$$

$$\frac{dm}{dt} = 0. \tag{6}$$

Drag opposes motion, while lift and gravity change the flight-path angle. Ablation is neglected, so mass remains constant.

The aerodynamic forces are

$$D = \frac{1}{2}\rho_a(h)V^2 C_D A, \qquad L = \frac{1}{2}\rho_a(h)V^2 C_L A. \tag{7}$$

The coefficients C_D and C_L are constant; Mach-number and angle-of-attack dependence are neglected. The forces still vary through their dependence on speed and atmospheric density.

Density and gravitational acceleration are modeled as

$$\rho_a(h) = \rho_{30}\exp\left[-\frac{h-h_0}{H}\right], \qquad g(h) = g_0\left(\frac{R_E}{R_E+h}\right)^2, \tag{8}$$

where $h_0 = 30000$ m is the fixed reference altitude. The exponential density approximation couples altitude to aerodynamic forces and heating.

B. Thermal Response

Convective heating is represented by a Sutton–Graves-type correlation [2],

$$\dot{q}_{\mathrm{conv}} = K\sqrt{\frac{\rho_a(h)}{R_n}}V^3. \tag{9}$$

This correlation estimates stagnation-point heating. Accordingly, T_s represents an approximate stagnation-region temperature rather than a uniform temperature over the aircraft.

An energy balance gives

$$\frac{dT_s}{dt} = \frac{\dot{q}_{\mathrm{conv}} - \epsilon\sigma T_s^4}{C_A}, \qquad C_A = \rho_s c_p \delta = 7200\ \mathrm{J/(m^2\,K)}. \tag{10}$$

Incoming environmental radiation, conduction into the vehicle interior, and material degradation are neglected. The effective carbon–carbon material properties are assumed constant. Temperature does not feed back into the aerodynamic forces or convective heating in this model.

C. Baseline Parameter Values

Table 2 specifies the baseline parameters. Vehicle geometry, aerodynamic coefficients, and material properties are assumed inputs rather than verified X-43 measurements. Each parameter is held constant within a run; only the selected study parameter changes between runs.

Table 2 Baseline model parameters.

Parameter	Value	Units
Drag coefficient, C_D	0.02	Dimensionless
Lift coefficient, C_L	0.080	Dimensionless
Reference area, A	4.00	$\mathrm{m^2}$
Nose radius, R_n	0.010	m
Surface emissivity, ϵ	0.85	Dimensionless
Specific heat, c_p	900	$\mathrm{J/(kg\,K)}$
Material density, ρ_s	1600	$\mathrm{kg/m^3}$
Heated thickness, δ	0.005	m
Reference air density, ρ_{30}	0.0184	$\mathrm{kg/m^3}$
Atmospheric scale height, H	7000	m
Surface gravity, g_0	9.81	$\mathrm{m/s^2}$
Earth radius, R_E	6.371×10^6	m
Heating coefficient, K	1.7415×10^{-4}	$\mathrm{kg^{1/2}/m}$
Stefan–Boltzmann constant, σ	$5.670374419 \times 10^{-8}$	$\mathrm{W/(m^2\,K^4)}$

IV. Numerical Method

A. Selection of RK4

The five-state IVP is integrated using the classical four-stage Runge–Kutta method presented in Lecture 10 [3], implemented from scratch. RK4 is explicit and requires four evaluations of the complete derivative function per timestep, without a nonlinear algebraic solve. For this small system, that cost makes repeated parameter studies practical. Its fourth-order accuracy also reduces the number of steps needed relative to a first-order method when both operate in their convergence regimes. Timestep refinement is used to assess accuracy for the simulated conditions.

B. RK4 Update

Starting from time t_k and state $\mathbf{y}_k$, the exact update satisfies

$$\mathbf{y}(t_k + \Delta t) = \mathbf{y}(t_k) + \int_{t_k}^{t_k+\Delta t} \mathbf{f}(t, \mathbf{y}(t))\, dt. \tag{11}$$

RK4 approximates this integral using four derivative estimates:

$$\mathbf{k}_1 = \mathbf{f}(t_k, \mathbf{y}_k), \tag{12}$$

$$\mathbf{k}_2 = \mathbf{f}\left(t_k + \frac{\Delta t}{2}, \mathbf{y}_k + \frac{\Delta t}{2}\mathbf{k}_1\right), \tag{13}$$

$$\mathbf{k}_3 = \mathbf{f}\left(t_k + \frac{\Delta t}{2}, \mathbf{y}_k + \frac{\Delta t}{2}\mathbf{k}_2\right), \tag{14}$$

$$\mathbf{k}_4 = \mathbf{f}(t_k + \Delta t, \mathbf{y}_k + \Delta t\mathbf{k}_3). \tag{15}$$

The state is advanced by

$$\mathbf{y}_{k+1} = \mathbf{y}_k + \frac{\Delta t}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4). \tag{16}$$

Each stage is a five-component vector. Density, gravity, aerodynamic forces, and heat fluxes are recalculated at that stage's intermediate state, so the entire coupled system is advanced together.

C. Accuracy and Truncation Error

Lecture 10 presents classical RK4 as a fourth-order one-step method [3]. Following Lecture 11, we examine accuracy by substituting the exact solution into the update and comparing Taylor expansions [4]. For a smooth exact solution and $s = \Delta t$,

$$\mathbf{y}(t_k + s) = \mathbf{y}(t_k) + s\mathbf{y}'(t_k) + \frac{s^2}{2}\mathbf{y}''(t_k) + \frac{s^3}{6}\mathbf{y}^{(3)}(t_k) + \frac{s^4}{24}\mathbf{y}^{(4)}(t_k) + O(s^5). \tag{17}$$

RK4 uses its two midpoint estimates and endpoint estimate to match these terms through fourth order. The matching can be shown directly for the scalar test equation $u' = \lambda u$. Starting from the exact value $u(t_k)$, the four stages are

$$k_1 = \lambda u(t_k), \tag{18}$$

$$k_2 = \lambda u(t_k)\left(1 + \frac{\lambda s}{2}\right), \tag{19}$$

$$k_3 = \lambda u(t_k)\left(1 + \frac{\lambda s}{2} + \frac{\lambda^2 s^2}{4}\right), \tag{20}$$

$$k_4 = \lambda u(t_k)\left(1 + \lambda s + \frac{\lambda^2 s^2}{2} + \frac{\lambda^3 s^3}{4}\right). \tag{21}$$

Substituting these expressions into the RK4 update gives

$$u_{k+1} = u(t_k)\left[1 + \lambda s + \frac{\lambda^2 s^2}{2} + \frac{\lambda^3 s^3}{6} + \frac{\lambda^4 s^4}{24}\right]. \tag{22}$$

The exact solution advances by $u(t_k+s) = u(t_k)e^{\lambda s}$, whose Taylor series has the same terms through s^4. The difference after one step is therefore $O(s^5)$.

Using Lecture 11's convention, the truncation error is this one-step residual divided by the timestep:

$$\tau_k = \frac{u(t_k + s) - u_{k+1}}{s} = O(s^4). \tag{23}$$

Thus the $O(s^5)$ one-step solution error and the $O(s^4)$ truncation error describe the same calculation using different conventions. For smooth problems with stable time integration, RK4 has global error $O(s^4)$ over a fixed interval. Halving the timestep should therefore reduce that error by approximately sixteen.

The scalar calculation illustrates the Taylor matching; it is not a proof of order for every nonlinear vector system. For the present coupled model, the timestep study tests the expected fourth-order behavior directly, and the exact drag and cooling cases independently check the computed solutions.

D. Linear Stability

Following Lecture 12, apply the numerical method to $u' = \lambda u$ and write the update as $u_{n+1} = R(w)u_n$, where $w = \lambda\Delta t$ [5]. The stage substitution above gives

$$R(w) = 1 + w + \frac{w^2}{2} + \frac{w^3}{6} + \frac{w^4}{24}. \tag{24}$$

For a decaying mode, $|R(w)| < 1$ means that its numerical amplitude decreases; $|R(w)| > 1$ means it grows. The boundary is $|R(w)| = 1$. On the negative real axis, RK4 is stable for $-2.78529 < w < 0$, consistent with the limit reported in Lecture 12.

Following the local linearization approach in Lectures 12–13 [5, 6], the temperature derivative gives the thermal eigenvalue

$$\lambda_T = -\frac{4\epsilon\sigma T_s^3}{C_A}. \tag{25}$$

Along the baseline trajectory, this gives a minimum local thermal stability limit of approximately 5.70 s. The selected timestep of 0.1 s is below this limit. This calculation assesses the thermal mode only; it does not establish stability of the complete nonlinear system. Full-model convergence and parameter-case refinement provide additional numerical evidence. At the production step, the baseline takes 4000 ordinary steps and 16000 RK4 stage evaluations. This cost is modest for five states, and no implicit equation solves are needed.

E. Algorithmic Implementation

At each step, the code evaluates the four stage vectors, forms the weighted update, stores the new state, and advances time. The last step is shortened when necessary to end at the requested final time. If a step crosses the specified altitude threshold, the code locates the crossing by repeatedly halving the bracketed time interval and recomputing a shortened RK4 step. This stopping calculation is not used in any reported case because all cases reach the time horizon first. Integration stops at that event or at the time horizon, whichever occurs first.

V. Implementation Verification and Timestep Selection

A. Timestep-Convergence Study

The baseline five-state IVP was integrated over $0 \le t \le 400$ s using timesteps of 1.0, 0.5, 0.25, and 0.125 s. The initial conditions and physical parameters were held fixed. A reference solution was computed using the same from-scratch RK4 implementation with $\Delta t_{\mathrm{ref}} = 0.03125$ s. All comparisons were made at matching time nodes, so interpolation was unnecessary.

For each state component j, the maximum error relative to the reference was defined as

$$E_j(\Delta t) = \max_n \left| y_j^{(\Delta t)}(t_n) - y_j^{(\mathrm{ref})}(t_n) \right|. \tag{26}$$

The observed convergence order was calculated from successive timestep refinements:

$$p_j = \log_2 \left(\frac{E_j(\Delta t)}{E_j(\Delta t/2)} \right). \tag{27}$$

For a fourth-order method, p_j should approach four, corresponding to an error reduction of approximately sixteen when the timestep is halved.

Table 3 Full-IVP convergence against an RK4 reference with $\Delta t_{\mathrm{ref}} = 0.03125$ s. The order in each row compares that row with the preceding row.

Δt (s)	E_h (m)	E_V (m/s)	E_{T_s} (K)	p_{T_s}
1.000	1.72162×10^{-4}	2.23630×10^{-6}	5.91707×10^{-2}	—
0.500	1.07599×10^{-5}	1.39611×10^{-7}	3.22222×10^{-3}	4.199
0.250	6.72269×10^{-7}	8.71842×10^{-9}	1.88395×10^{-4}	4.096
0.125	4.18695×10^{-8}	5.52291×10^{-10}	1.13451×10^{-5}	4.054

Table 3 shows that the observed temperature order approaches four as the timestep decreases. Altitude and velocity errors also decrease by approximately a factor of sixteen with each refinement. These results demonstrate the expected fourth-order behavior for the coupled flight and thermal calculation.

Because the reference solution is numerical, this study establishes self-convergence rather than an independent verification of the governing equations. The exact special-case solutions, derivative benchmarks, and mechanical energy check below address that separate requirement.

B. Independent Verification Against Exact Special Cases

Self-convergence cannot show that the correct equations have been solved. Two analytically solvable limits were therefore integrated using the same five-state model function and RK4 routine as the full problem. Only parameters and initial conditions were changed; no separate simplified numerical right-hand side was used.

1. Constant-altitude drag

Set $g_0 = 0$, $C_L = 0$, and $\gamma_0 = 0$, and disable convective heating and radiation. Then h, γ, T_s, and m remain constant, and density is fixed at $\rho_a(h_0)$. The velocity equation reduces to

$$\frac{dV}{dt} = -aV^2, \qquad a = \frac{\rho_a(h_0) C_D A}{2m_0}. \tag{28}$$

Integrating $dV/V^2 = -a\,dt$ gives

$$V_{\mathrm{exact}}(t) = \frac{V_0}{1 + aV_0 t}. \tag{29}$$

This test uses $h_0 = 30000$ m, $V_0 = 2112.14$ m/s, $T_{s,0} = 300$ K, $m_0 = 1400$ kg, and a 20 s interval. For this verification experiment only, $C_D = 2$ makes the truncation error measurable above floating-point roundoff. Production simulations retain $C_D = 0.02$.

Table 4 Convergence to the exact constant-altitude drag solution. Error is the maximum speed difference at sampled time nodes.

Δt (s)	Maximum error (m/s)	Observed order
2.000	1.97324×10^{-2}	—
1.000	1.40536×10^{-3}	3.812
0.500	8.99864×10^{-5}	3.965
0.250	5.66429×10^{-6}	3.990
0.125	3.54472×10^{-7}	3.998

Table 4 shows convergence to an independently known solution, with order approaching four. This checks the drag sign, coefficient and mass dependence, and its time integration in this limit.

2. Radiative cooling

Set atmospheric density and gravity to zero and choose $\gamma_0 = 0$. The flight variables remain constant, convective heating vanishes, and

$$\frac{dT_s}{dt} = -\frac{\epsilon\sigma}{C_A} T_s^4. \tag{30}$$

Integrating $T_s^{-4} dT_s = -(\epsilon\sigma/C_A)dt$ yields

$$T_{s,\mathrm{exact}}(t) = \left(T_{s,0}^{-3} + \frac{3\epsilon\sigma}{C_A} t \right)^{-1/3}. \tag{31}$$

Using $T_{s,0} = 2500$ K, $\epsilon = 0.85$, and $C_A = 7200$ J/(m^2 K) over 20 s gives Table 5. Other initial flight variables retain their baseline values except for $\gamma_0 = 0$.

Table 5 Convergence to the exact radiative-cooling solution. Errors are maximum temperature differences at sampled time nodes.

Δt (s)	Maximum error (K)	Observed order
0.25000	7.19051×10^{-6}	—
0.12500	2.87742×10^{-7}	4.643
0.06250	1.32186×10^{-8}	4.444
0.03125	6.81212×10^{-10}	4.278

The errors decrease toward the exact cooling solution, checking the radiation sign, fourth-power temperature dependence, and heat-capacity factor. The apparent orders above four at these steps approach four under refinement; they do not imply a higher formal order for RK4.

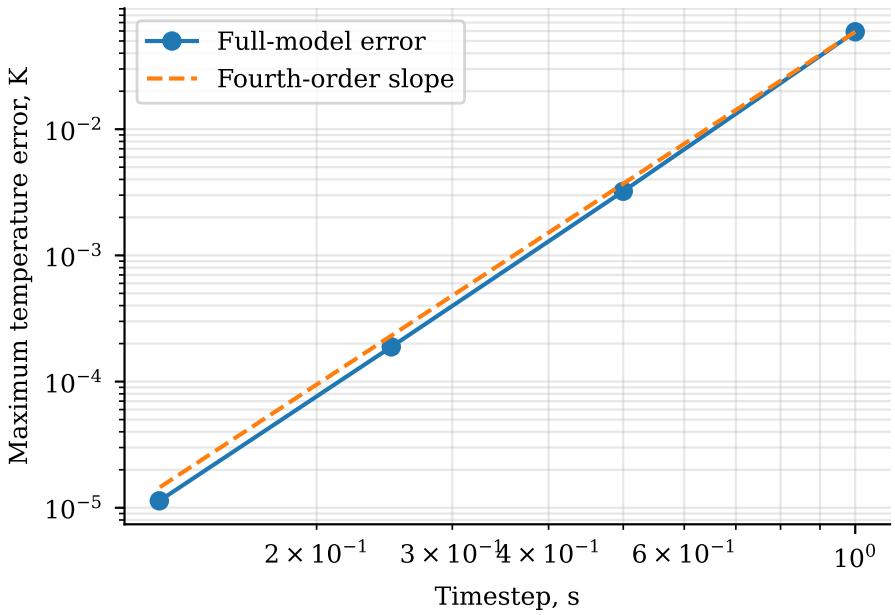


Fig. 1 Full-model maximum temperature error under timestep refinement, compared with a fourth-order slope. Independent exact-solution results are reported in Tables 4 and 5.

C. Derivative Benchmarks and Mechanical Energy Check

To complement the exact limits, the complete derivative function was checked against independent calculations at two prescribed operating states.

Two prescribed states were selected:

$$\mathbf{y}_A = [30000,\ 2112.14,\ -2\pi/180,\ 300,\ 1400]^T, \tag{32}$$

$$\mathbf{y}_B = [20000,\ 1800,\ -4\pi/180,\ 2200,\ 1400]^T. \tag{33}$$

The state components use meters, meters per second, radians, kelvin, and kilograms, respectively. State A represents the baseline initial condition. State B checks the equations at a lower altitude and higher surface temperature, where atmospheric density and radiative cooling differ substantially.

For each state, density and gravity were evaluated independently. Drag, lift, and both heat fluxes were then calculated using scalar arithmetic without the project's model helper functions. Substitution into the five governing equations gave the benchmark derivatives.

The implemented derivative function matched the unrounded benchmark values using relative and absolute tolerances of 10^{-12}. This comparison checks the signs, coefficients, and state dependence of the implemented equations at two distinct operating conditions.

An additional check used an independently derived mechanical energy identity. Define the

Table 6 Independent derivative benchmarks. Values shown are rounded for presentation.

Derivative	State A	State B
$\dot{h}$ (m/s)	−73.71262297	−125.56165274
$\dot{V}$ (m/s^2)	−2.00612018	−6.42744249
$\dot{\gamma}$ (rad/s)	$-1.56812807 \times 10^{-4}$	$1.03916439 \times 10^{-2}$
$\dot{T}_s$ (K/s)	309.09488520	234.05040846
$\dot{m}$ (kg/s)	0	0

specific mechanical energy as

$$E = \frac{V^2}{2} + \frac{g_0 R_E h}{R_E + h}. \tag{34}$$

The derivative of the potential-energy term with respect to altitude is $g(h)$. Therefore,

$$\frac{dE}{dt} = V\frac{dV}{dt} + g(h)\frac{dh}{dt}. \tag{35}$$

Substituting the governing equations gives

$$\frac{dE}{dt} = V\left(-\frac{D}{m} - g\sin\gamma\right) + gV\sin\gamma = -\frac{DV}{m}. \tag{36}$$

Thus, gravity transfers energy between kinetic and potential forms, while drag removes mechanical energy. The implemented derivatives satisfied this identity at the baseline initial state to a relative tolerance of 10^{-13}.

Together, the exact special cases, independent derivative benchmarks, and mechanical energy check provide evidence that the governing equations were translated into code correctly. The timestep-convergence study separately demonstrates the expected fourth-order behavior of RK4. These checks verify numerical implementation; they do not validate the physical assumptions of the model.

D. Selection of the Production Timestep

A production timestep of 0.1 s was selected. Each of the 22 parameter cases was repeated using 0.05 s. The acceptance criteria were a peak-temperature change below 0.5 K and relative changes below 0.1% in both peak convective heat flux and integrated convective heat load.

All cases satisfied these criteria. The largest change in peak temperature was 6.53429×10^{-4} K. Together with the convergence study, these results support the selected timestep for the reported studies. The largest relative changes in peak flux and integrated heat load are approximately 4.84×10^{-7} and 4.40×10^{-8}, respectively. These differences are empirical accuracy indicators rather

than rigorous error bounds. Peak differences include sampling effects; heat load additionally has trapezoidal quadrature error.

VI. Results

Each simulation ends at 400 s or the first downward crossing of 10 km. All reported cases reached the time horizon first, so the results describe a common observation interval rather than a complete descent. The parameter studies vary one quantity at a time: initial speed from 1700 to 2500 m/s in 100 m/s increments, initial flight-path angle from -6° to 0° in 1° increments, and emissivity from 0.5 to 1.0 in increments of 0.1. Other inputs retain their baseline values.

In addition to peak temperature and peak flux, the integrated convective heat load per unit area is

$$Q_{\mathrm{conv}} = \int_0^{t_{\mathrm{end}}} \dot{q}_{\mathrm{conv}}(t)\, dt, \tag{37}$$

evaluated using the composite trapezoidal rule. Reported peaks are maxima sampled at the numerical time nodes; the step-halving study checks the sensitivity of these quantities to temporal resolution.

A. Baseline Simulation

Using the baseline state and parameters with $\Delta t = 0.1$ s, the temperature reaches 2632.82 K at approximately 28 s. Peak convective heat flux is 2.317 MW/m^2, and integrated convective heat load is 563.57 MJ/m^2. At 400 s, altitude is 21.34 km and speed is 1178.83 m/s.

Figure 2 shows the trajectory and thermal histories. The altitude oscillations arise because lift changes the flight-path angle. During descent, increasing density can raise lift above $mg\cos\gamma$, turning the trajectory upward. As the vehicle climbs, density and lift decrease, allowing gravity to turn the trajectory downward again. Drag removes mechanical energy throughout, producing an overall descent with temporary climbs. This behavior is a prediction of the simplified equations rather than a validated X-43 flight characteristic.

B. Effect of Initial Velocity

As shown in Fig. 3, increasing initial speed from 1700 to 2500 m/s increases peak convective heat flux from 1.532 to 3.732 MW/m^2 and peak temperature from 2373.43 to 2962.27 K. This trend is consistent with the V^3 dependence of convective heating. The peak values also depend on how the changed initial speed modifies the subsequent density and trajectory histories, so they cannot be inferred from the initial-speed ratio alone. Integrated heat load also increases from 335.56 to 837.50 MJ/m^2 over the observation interval. Thus both the instantaneous heating demand and

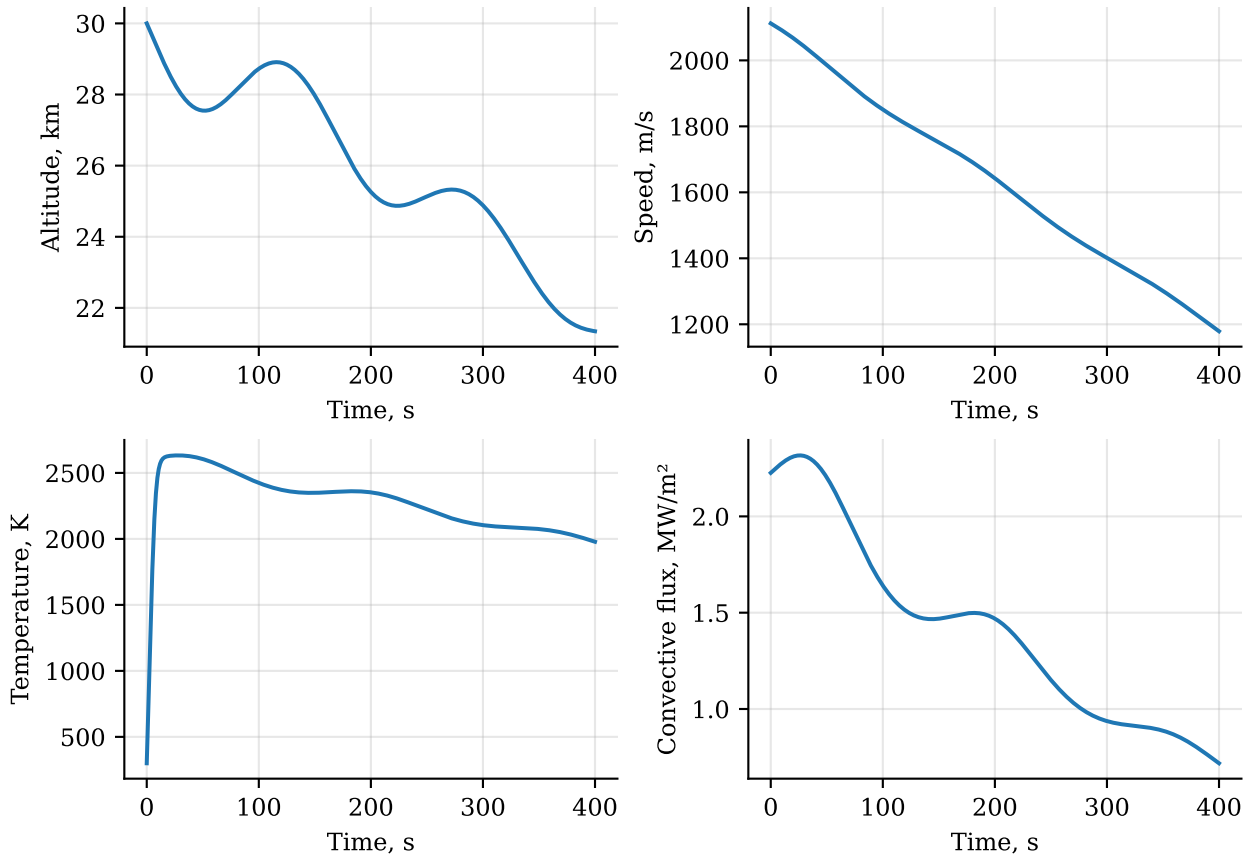


Fig. 2 Baseline altitude, speed, approximate stagnation-region temperature, and convective heat flux over the 400 s **observation interval.**

cumulative exposure increase with initial speed over this range.

C. Effect of Surface Emissivity

Figure 4 shows that increasing emissivity from 0.5 to 1.0 reduces peak temperature from 3006.14 to 2528.01 K, while peak convective heat flux remains 2.317 MW/m^2. Higher emissivity increases radiative heat loss. Because temperature does not feed back into the flight equations, the trajectory and incoming convective heat flux are unchanged in this study. The integrated incoming heat load remains 563.57 MJ/m^2. Emissivity controls outgoing radiation while leaving incoming heating unchanged.

D. Effect of Initial Flight-Path Angle

The supplementary angle study in Fig. 5 shows that changing the initial angle from 0° to −6° raises peak temperature from 2572.17 to 2777.78 K and peak convective heat flux from 2.226 to 2.873 MW/m^2. However, integrated heat load decreases from 566.00 to 549.11 MJ/m^2 over the

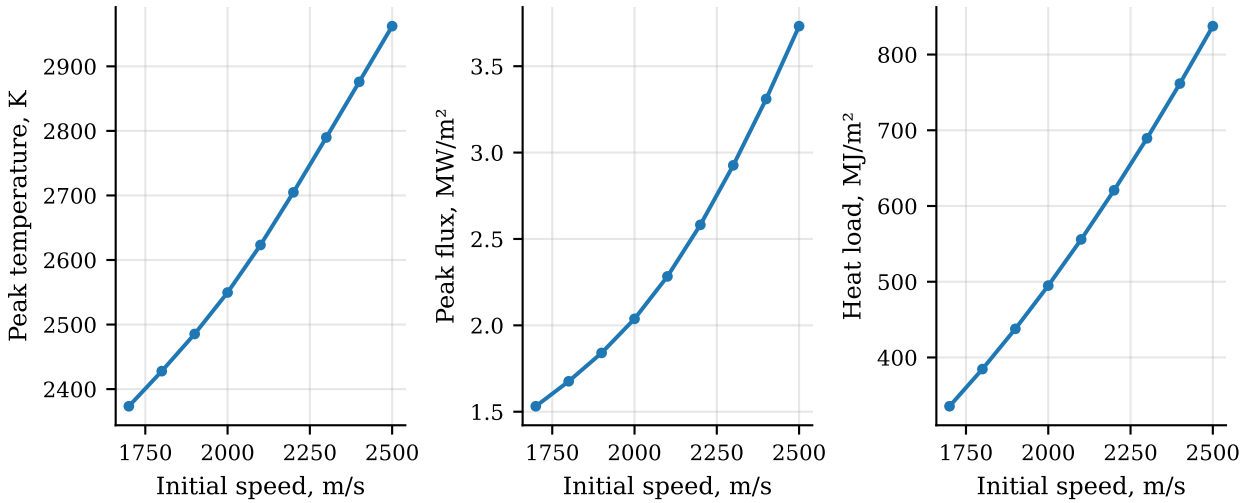


Fig. 3 Effect of initial speed on peak temperature, peak convective flux, and integrated heat load. Other inputs retain baseline values.

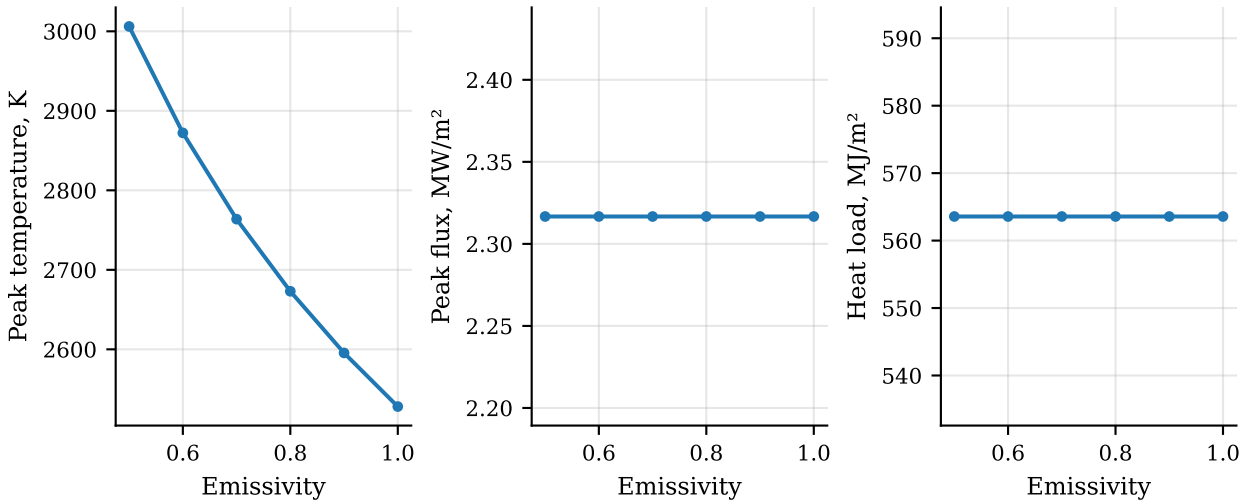


Fig. 4 Effect of emissivity on peak temperature, peak convective flux, and integrated heat load. The trajectory and incoming heating are unchanged.

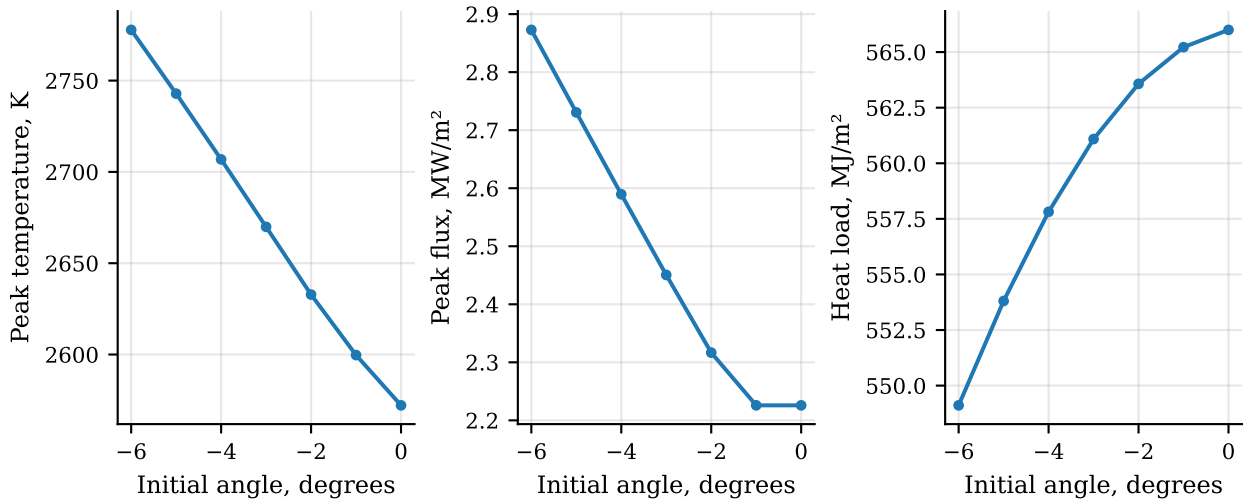


Fig. 5 Effect of initial flight-path angle on peak temperature, peak convective flux, and integrated heat load. Other inputs retain baseline values.

fixed observation interval. Peak heating and cumulative exposure therefore need not change in the same direction when the trajectory changes.

E. Model Limitations

The computations verify numerical behavior under the stated equations, not predictive fidelity to flight data. Constant lift and drag coefficients omit Mach-number and angle-of-attack effects; the exponential atmosphere omits temperature-dependent atmospheric structure. The locally flat flight geometry neglects curvature even though the calculation spans several hundred seconds. These assumptions can affect the trajectory and heating history. The stagnation-point correlation is used as an effective local heating model, not a resolved heat-transfer prediction over a slender aircraft.

The single thermal state excludes internal conduction, oxidation, ablation, and temperature-dependent material properties. Consequently, temperatures near 2500–3000 K cannot be interpreted as proof that the assumed material survives. No allowable-temperature or failure criterion is specified. A more predictive model would require validated aerodynamic data, a suitable atmospheric model, curved-flight geometry, and a calibrated thermal-protection response. The present trends describe the selected 400 s interval; extrapolation to landing or a complete entry is outside the study.

VII. Conclusions

A from-scratch RK4 implementation was used to study a coupled flight and thermal IVP. Full-model timestep refinement approached fourth-order accuracy, and convergence to exact drag and

radiative-cooling limits supplied independent solution-level checks. Derivative benchmarks and a mechanical energy identity provided additional implementation evidence. A production step of 0.1 s was supported by refinement of all 22 parameter cases.

For the assumed model, higher initial speed increased peak convective heating, peak temperature, and integrated heat load. Greater emissivity lowered peak temperature without changing incoming convective heating or the trajectory. A steeper initial descent increased peak heating but slightly reduced heat load over the fixed observation window, showing why both peak and cumulative metrics matter. The baseline peaked at 2632.82 K and remained above 10 km at 400 s. These results answer the parameter questions within the simplified model rather than establishing vehicle thermal safety or actual X-43 performance.

Organization of Work

PLACEHOLDER: Insert each group member's actual technical roles and approximate percentage contribution.

Reproducibility

The development repository is `https://github.com/AkonQiu/AE370-Group`. The accompanying `AE370-submission.zip` contains a snapshot of the code, generated data, and report source used for this document. The implementation separates the model and RK4 solver in `src/`, verification and parameter studies in `analysis/`, automated checks in `tests/`, and generated data and plots in `results/`. The report source is `report/main.tex`. From the extracted repository root, with Python 3.11 or later, run:

```
python -m venv .venv
.venv\Scripts\python -m pip install -r requirements.txt
.venv\Scripts\python -m pytest -q
.venv\Scripts\python -m analysis.run_all
.venv\Scripts\python -m analysis.report_figures
```

These commands use Windows paths. On Linux or macOS, substitute `.venv/bin/python` for `.venv\Scripts\python`. No outside time integrator is called to advance the project solutions. The analysis regenerates the baseline history, exact-limit and full-model convergence tables, all 22 parameter cases, refinement comparisons, and figures. Configuration values and software versions are recorded in `results/summary.json`. The recorded analysis environment used Python 3.14.7, NumPy 2.5.2, and Matplotlib 3.11.1. The current automated test suite contains 31 passing tests,

including parametrized cases. The final command generates figures at their intended report size. The report folder includes its figure files and build instructions.

Acknowledgments

OpenAI ChatGPT and Codex assisted with developing and organizing the premise, discussing RK4 selection and verification, reviewing and extending the initial code, implementing parameter studies and automated checks, generating plots, and drafting and revising the report. This assistance included mathematical explanations and interpretation of numerical results. The model parameters and original governing equations were supplied in the group's project premise, and the initial repository already contained an RK4 implementation and baseline flight model. AI assistance does not constitute independent experimental validation of the model assumptions.

References

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[2] Sutton, K., and Graves, R. A., Jr., "A General Stagnation-Point Convective-Heating Equation for Arbitrary Gas Mixtures," NASA TR R-376, November 1971. `https://ntrs.nasa.gov/citations/19720003329`.

[3] Goza, A., "Lecture 10: Initial Value Problems and One-Step Methods," AE 370: Aerospace Numerical Methods, Fall 2026, course lecture slides.

[4] Goza, A., "Lecture 11: Multi-Step Methods and Truncation Error," AE 370: Aerospace Numerical Methods, Fall 2026, course lecture slides.

[5] Goza, A., "Lecture 12: Absolute Stability," AE 370: Aerospace Numerical Methods, Fall 2026, course lecture slides.

[6] Goza, A., "Lecture 13: Stiff Initial Value Problems," AE 370: Aerospace Numerical Methods, Fall 2026, course lecture slides.